Project Report

Python-Based Projectile Motion Calculator

Project Type: Independent Programming Project
Programming Language: Python
Subject Areas: Physics, Mathematics and Computer Science

1. Introduction

Projectile motion is a type of motion in which an object is launched into the air and moves under the influence of gravity. It is an important topic in Physics and is used to understand the movement of objects such as balls, stones and other projectiles.

I chose this project because I wanted to combine my knowledge of A-level Physics with my interest in programming. My aim was to create a simple Python programme that could calculate important values related to projectile motion.

2. Objective

The main objective of this project is to develop a Python-based calculator that can calculate:

- The time of flight of a projectile.
- The maximum height reached by the projectile.
- The horizontal range of the projectile.

The programme takes the initial velocity and projection angle as inputs from the user and uses mathematical equations to calculate the results.

3. Physics and Mathematical Principles

The calculator uses the equations of projectile motion.

When an object is projected at an angle, its initial velocity can be divided into two components:

Horizontal velocity:

(Ux = ucos\theta)

Vertical velocity:

(Uy = usin\theta)

Where:

- (U) = Initial velocity
- (\theta) = Angle of projection

The programme uses the following equations:

Time of flight:

$$\frac{2usin/theta}{g}$$

Maximum height:

$$\frac{u^2sin^2/theta}{2g}$$

Horizontal range:

$$\frac{u^2sin2/theta}{g}$$

Here, g represents gravitational acceleration, which is taken as 9.81 m/s^2.

The calculations assume that air resistance is ignored and that the projectile lands at the same height from which it was launched.

4. Programming and Implementation

I used Python to develop this calculator.

The programme asks the user to enter an initial velocity and an angle of projection. It then converts the angle from degrees into radians and performs the necessary calculations.

Some of the Python concepts used in the programme include:

- Variables to store numerical values.
- Input statements to receive information from the user.

- Data type conversion to handle decimal numbers.
- Mathematical operators to perform calculations.
- Functions to organise the programme.
- Conditional statements to check user input.

The programme then displays the calculated time of flight, maximum height and horizontal range.

5. Testing and Results

To test the programme, different values of initial velocity and projection angle can be entered.

For example, consider a projectile launched with an initial velocity of 20 m/s at an angle of 45°.

The expected results are:

Quantity	Calculated result
Initial velocity	20 m/s
Projection angle	45°
Time of flight	2.88 s
Maximum height	10.19 m
Horizontal range	40.77 m

These results can be compared with manual calculations to check whether the programme is working correctly.

6. Challenges and Learning Experience

One of the main challenges was understanding how to convert the Physics equations into Python calculations.

I also had to understand how angles are handled in Python's mathematical functions, as these functions generally require angles in radians rather than degrees.

Through this project, I gained a better understanding of how programming can be used to solve mathematical and scientific problems. It also helped me connect the concepts I study in A-level Physics with practical applications.

7. Limitations and Future Improvements

The current calculator uses a simplified model of projectile motion. It ignores air resistance and assumes that the projectile lands at the same height from which it was launched.

In the future, the programme could be improved by adding a graph to display the projectile's trajectory. It could also include air resistance and allow users to compare different launch angles and initial velocities.

8. Conclusion

This project allowed me to apply my knowledge of Physics and Mathematics using Python. I learned how mathematical equations can be converted into a working computer programme.

Although the calculator is relatively simple, it helped me develop my programming skills and explore the connection between Computer Science and Engineering. It has also encouraged me to explore more engineering-related programming projects in the future.